

Multivariable Calculus: A Complete Study Guide

From Limits to Multiple Integrals — Explained Simply

PART I: LIMITS IN MULTIVARIABLE CALCULUS

1. Limits of Functions of Two and Three Variables

What Is a Limit in 2D?

For a single-variable function, the limit $(\lim_{x \rightarrow a} f(x) = L)$ means: as (x) gets closer to (a) , $(f(x))$ gets closer to (L) .

For two variables, the limit $(\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L)$ means: as the point $((x, y))$ approaches $((a, b))$ from

any direction

, $(f(x,y))$ must approach the same value (L) .

Key Insight: In single-variable calculus, you can only approach from the left or right. In two variables, you can approach from **infinitely many directions** — along lines, curves, spirals, etc. This makes limits much harder to prove!

Example — A Simple Limit

$$[\lim_{(x,y) \rightarrow (1,2)} (3x + 2y) = 3(1) + 2(2) = 7]$$

For polynomial and rational functions (where denominator $\neq 0$ at the point), you can just substitute directly.

Limits of Three-Variable Functions

For $(f(x, y, z))$, the same idea applies:

$$[\lim_{(x,y,z) \rightarrow (a,b,c)} f(x,y,z) = L]$$

The point $((x, y, z))$ approaches $((a, b, c))$ from all directions in 3D space.

2. Showing a Limit Does NOT Exist

If a function approaches **different values** along different paths to the same point, the limit **does not exist**. Three standard methods:

Method 1: The Two-Path Test

Strategy:

Find two different paths to the point that give different limit values.

Example:

Show $(\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2})$ does not exist.

Path 1:

Along the x-axis, set $(y = 0)$: $[\frac{x^2 - 0}{x^2 + 0} = \frac{x^2}{x^2} = 1 \quad \rightarrow \quad \text{limit} = 1]$

Path 2:

Along the y-axis, set $(x = 0)$: $[\frac{0 - y^2}{0 + y^2} = \frac{-y^2}{y^2} = -1 \quad \rightarrow \quad \text{limit} = -1]$

Since the two paths give **different values** ($1 \neq -1$), the limit **does not exist**.

Remember: You only need to find **two paths** that give different answers. But if all paths give the same answer, that alone does not prove the limit exists — you need polar coordinates or the squeeze theorem for that.

Method 2: Repeated (Iterated) Limits

Iterated limits evaluate the limit one variable at a time:

$$\left[\lim_{x \rightarrow a} \left[\lim_{y \rightarrow b} f(x,y) \right] \right] \quad \text{vs.} \quad \lim_{y \rightarrow b} \left[\lim_{x \rightarrow a} f(x,y) \right]$$

Rule: If the two iterated limits exist but are **not equal**, then the simultaneous limit does not exist.

Example:

$$(f(x,y) = \frac{x - y}{x + y}) \text{ as } (x,y) \rightarrow (0,0)$$

$$\text{First iterated limit (fix } (x), \text{ let } (y \rightarrow 0) \text{ first): } \left[\lim_{x \rightarrow 0} \left[\lim_{y \rightarrow 0} \frac{x - y}{x + y} \right] \right] = \lim_{x \rightarrow 0} \frac{x}{x} = \lim_{x \rightarrow 0} 1 = 1$$

$$\text{Second iterated limit (fix } (y), \text{ let } (x \rightarrow 0) \text{ first): } \left[\lim_{y \rightarrow 0} \left[\lim_{x \rightarrow 0} \frac{x - y}{x + y} \right] \right] = \lim_{y \rightarrow 0} \frac{-y}{y} = \lim_{y \rightarrow 0} (-1) = -1$$

Since $1 \neq -1$, the double limit does not exist.

Method 3: Polar Coordinates

Convert to polar form using $(x = r\cos\theta, y = r\sin\theta)$. As $(x,y) \rightarrow (0,0)$, we have $(r \rightarrow 0)$.

If the result depends on (θ)

, the limit does not exist.

Example:

$$(f(x,y) = \frac{xy}{x^2 + y^2})$$

$$\text{Substitute: } \left[f = \frac{r\cos\theta \cdot r\sin\theta}{r^2} = \frac{r^2 \cos\theta \sin\theta}{r^2} = \cos\theta \sin\theta \right]$$

As $(r \rightarrow 0)$, the result is $(\cos\theta \sin\theta)$, which

depends on the angle (θ)

. So the limit does not exist.

Example of a limit that DOES exist:

$$(g(x,y) = \frac{x^2 y}{x^2 + y^2})$$

$$\left[g = \frac{r^2 \cos^2\theta \cdot r\sin\theta}{r^2} = r\cos^2\theta \sin\theta \right]$$

As $(r \rightarrow 0)$, this $(\rightarrow 0)$ regardless of (θ) . So $(\lim_{(x,y) \rightarrow (0,0)} g = 0)$. ✓

3. Continuity of Multivariable Functions

A function $f(x,y)$ is

continuous at

$((a,b))$ if all three conditions hold:

1. $f(a,b)$ is defined (the function exists at the point)
2. $(\lim_{(x,y) \rightarrow (a,b)} f(x,y))$ exists
3. $(\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b))$ (limit equals the function value)

Simple Rule:

If you can plug in the values directly and get a defined answer, the function is continuous there. Polynomials are always continuous. Rational functions are continuous everywhere except where the denominator is zero.

Example:

$$f(x,y) = \frac{x^2 + y^2}{x - y}$$

This is continuous everywhere

except

on the line $(x = y)$, where the denominator is zero.

Continuity at a Boundary — Piecewise Functions

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Using polar coordinates, we showed $(\lim f = 0 = f(0,0))$. So this function

is continuous

at the origin.

PART II: PARTIAL DERIVATIVES AND DIFFERENTIALS

4. Partial Derivatives

The Core Idea

A partial derivative measures how a function changes when you change **one variable at a time**, keeping all others **frozen (constant)**.

For $(f(x, y))$: - $(\frac{\partial f}{\partial x})$ = differentiate with respect to (x) , treating (y) as a constant - $(\frac{\partial f}{\partial y})$ = differentiate with respect to (y) , treating (x) as a constant

Notation: (f_x) , $(\frac{\partial f}{\partial x})$, $(\partial_x f)$ — all mean the same thing.

Example:

$$(f(x,y) = x^3 y + 2x^2 y^2 - y^3)$$

$$[f_x = 3x^2 y + 4xy^2 \quad (\text{treat } y \text{ as constant})] [f_y = x^3 + 4x^2 y - 3y^2 \quad (\text{treat } x \text{ as constant})]$$

Higher-Order Partial Derivatives

You can differentiate again: - $(f_{xx}) = \frac{\partial^2 f}{\partial x^2}$ — differentiate (f_x) with respect to (x) again - $(f_{xy}) = \frac{\partial^2 f}{\partial x \partial y}$ — differentiate (f_x) with respect to (y) - $(f_{yx}) = \frac{\partial^2 f}{\partial y \partial x}$ — differentiate (f_y) with respect to (x)

Clairaut's (Mixed Partial) Theorem:

If (f_{xy}) and (f_{yx}) are continuous, then: $[f_{xy} = f_{yx}]$

The order of differentiation doesn't matter!

Geometric Meaning

If $(z = f(x,y))$ describes a surface: - $(f_x(a,b))$ = slope of the surface in the x -direction at point $((a,b))$ - $(f_y(a,b))$ = slope of the surface in the y -direction at point $((a,b))$

5. Total Differentials

What Is a Differential?

The
total differential

(df) approximates how much (f) changes when both (x) and (y) change by small amounts (dx) and (dy):

$$[df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy]$$

For three variables (f(x,y,z)):

$$[df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz]$$

Think of it this way:

Each partial derivative tells you how sensitive (f) is to that variable. The total differential adds up all the individual contributions.

Application: Error Estimation

The total differential is used to estimate **propagation of errors** in measurements.

Example:

The volume of a cylinder is ($V = \pi r^2 h$).

$$[dV = \frac{\partial V}{\partial r} dr + \frac{\partial V}{\partial h} dh = 2\pi r h dr + \pi r^2 dh]$$

If ($r = 5$) cm, ($h = 10$) cm, and measurement errors are ($dr = 0.1$) cm, ($dh = 0.2$) cm:

$$[dV = 2\pi(5)(10)(0.1) + \pi(25)(0.2) = 10\pi + 5\pi = 15\pi \approx 47.1 \text{ cm}^3]$$

6. Laplace's Equation

What Is It?

Laplace's equation describes

steady-state

phenomena — situations where nothing is changing over time. It appears in heat flow, electrostatics, fluid mechanics, and more.

In two variables: $[\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0]$

In three variables: $\left[\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0 \right]$

Written compactly: $(\nabla^2 f = 0)$

Example:

Verify $(f(x,y) = e^x \sin y)$ satisfies Laplace's equation.

$[f_{xx} = e^x \sin y, \quad f_{yy} = -e^x \sin y] [f_{xx} + f_{yy} = e^x \sin y - e^x \sin y = 0 \text{ \checkmark}]$

7. The Wave Equation

The wave equation models

vibrating strings, sound waves, light

, etc.

In one spatial dimension: $\left[\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \right]$

where (c) is the wave speed and $(u(x,t))$ is the displacement.

Example:

Verify $(u = \sin(x - ct))$ satisfies the wave equation.

$[u_{tt} = -c^2 \sin(x - ct), \quad u_{xx} = -\sin(x - ct)] [u_{tt} = c^2 u_{xx} \text{ \checkmark}]$

Physical Meaning:

The term (u_{tt}) is acceleration (how fast velocity changes), and (u_{xx}) measures the "curvature" of the wave shape. The wave equation says acceleration is proportional to curvature.

8. Harmonic Functions (Potential Functions)

A function is

harmonic

if it satisfies Laplace's equation $(\nabla^2 f = 0)$.

These are also called

potential functions

in physics — the gravitational or electric potential in free space is harmonic.

Properties of harmonic functions: - They are infinitely differentiable (very smooth) - They satisfy the **Mean Value Property**: the value at any point equals the average value on any circle/sphere around it - They cannot have local maxima or minima in the interior of their domain (only on boundaries)

Common harmonic functions:

$(f = x^2 - y^2)$	$(f_{xx} + f_{yy} = 2 + (-2) = 0) \checkmark$
$(f = e^x \cos y)$	$(f_{xx} + f_{yy} = e^x \cos y - e^x \cos y = 0) \checkmark$
$(f = \ln(x^2 + y^2))$	Check: $(f_{xx} + f_{yy} = 0) \checkmark$ (away from origin)

9. Linearization (Linear Approximation)

The Idea

Near a point $((a,b))$, a smooth function $(f(x,y))$ looks like a flat plane — its **tangent plane**. The equation of this tangent plane is the **linearization** $(L(x,y))$:

$$[L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)]$$

This is the 2D version of the single-variable tangent line approximation.

Example:

Linearize $(f(x,y) = \sqrt{x^2 + y^2})$ near $((3, 4))$.

$$\begin{aligned} [f(3,4) &= \sqrt{9+16} = 5] [f_x = \frac{x}{\sqrt{x^2+y^2}} \rightarrow f_x(3,4) = \frac{3}{5}] [f_y = \frac{y}{\sqrt{x^2+y^2}} \rightarrow f_y(3,4) = \frac{4}{5}] \\ [L(x,y) &= 5 + \frac{3}{5}(x-3) + \frac{4}{5}(y-4)] \end{aligned}$$

$$\text{So } f(3.1, 4.1) \approx 5 + \frac{3}{5}(0.1) + \frac{4}{5}(0.1) = 5 + 0.06 + 0.08 = 5.14$$

(Actual value: (≈ 5.141)) — Very accurate!

10. The Chain Rule

For Functions of Two Independent Variables

If $(w = f(x,y))$ and $(x = x(s,t))$, $(y = y(s,t))$, then:

$$\left[\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} \right]$$

$$\left[\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} \right]$$

Memory Aid:

Think of it as "all paths from w to the independent variable." Each path multiplies the derivatives along it, and you add all paths together.

Tree Diagram Method

Draw a tree: (w) branches to (x) and (y) . Each of those branches to (s) and (t) . To find $(\partial w / \partial s)$, trace all paths from (w) to (s) , multiply derivatives along each path, then add them.

For Three Independent Variables

If $(w = f(x,y,z))$ and each of (x, y, z) depends on (s, t, r) :

$$\left[\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s} \right]$$

Example:

$(w = x^2 + y^2)$, $(x = r \cos \theta)$, $(y = r \sin \theta)$

$$\left[\frac{\partial w}{\partial r} = 2x \cos \theta + 2y \sin \theta = 2r \cos^2 \theta + 2r \sin^2 \theta = 2r \right]$$

(This confirms $(w = r^2)$ in polar form — consistent!)

11. Implicit Differentiation

When Equations Define Functions Implicitly

Sometimes (z) is defined implicitly by $(F(x, y, z) = 0)$. You can find partial derivatives without solving for (z) explicitly:

$$\left[\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} \right]$$

Example:

Find $(\partial z / \partial x)$ if $(x^3 + y^3 + z^3 - 3xyz = 0)$.

Let $(F = x^3 + y^3 + z^3 - 3xyz)$.

$$[F_x = 3x^2 - 3yz, \quad F_z = 3z^2 - 3xy]$$

$$\left[\frac{\partial z}{\partial x} = -\frac{3x^2 - 3yz}{3z^2 - 3xy} = -\frac{x^2 - yz}{z^2 - xy} \right]$$

12. Partial Derivatives with Constrained Variables

The Problem

Sometimes variables are not all independent — they satisfy a constraint. For example, if $(w = f(x,y,z))$ and the variables satisfy $(g(x,y,z) = c)$, you want $(\left(\frac{\partial w}{\partial x}\right)_{\{z\}})$ (meaning: change (x) , keep (z) constant, let (y) adjust to satisfy the constraint).

Notation:

$(\left(\frac{\partial f}{\partial x}\right)_{\{y,z\}})$ means "partial derivative of (f) with respect to (x) , holding (y) and (z) fixed."

Strategy:

1. Use the constraint to express the dependent variable in terms of the independent ones
2. Substitute and differentiate

Example:

$(w = x^2 + y^2 + z^2)$ with constraint $(z = x^2 + y^2)$. Find $(\left(\frac{\partial w}{\partial x}\right)_y)$.

$$\text{Since } (z = x^2 + y^2): [w = x^2 + y^2 + (x^2 + y^2)^2] \left[\left(\frac{\partial w}{\partial x}\right)_y = 2x + 2(x^2 + y^2)(2x) = 2x + 4x(x^2 + y^2) = 2x(1 + 2z) \right]$$

13. Directional Derivatives

Motivation

Partial derivatives measure slopes only in the x or y directions. What about **any** direction?

The

directional derivative

of (f) in the direction of unit vector ($\mathbf{u} = \langle a, b \rangle$) (where ($a^2 + b^2 = 1$)) is:

$$[D_{\mathbf{u}} f(x,y) = f_x \cdot a + f_y \cdot b = \nabla f \cdot \mathbf{u}]$$

Important: ($\mathbf{u}$) must be a **unit vector**. Always normalize first:
($\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|}$).

Example:

Find the directional derivative of ($f(x,y) = x^2 y$) at ((1,2)) in the direction of ($\mathbf{v} = \langle 3, 4 \rangle$).

Unit vector: ($\mathbf{u} = \langle 3/5, 4/5 \rangle$)

$$[f_x = 2xy = 4, \quad f_y = x^2 = 1 \quad \text{at } (1,2)]$$

$$[D_{\mathbf{u}} f = 4 \cdot \frac{3}{5} + 1 \cdot \frac{4}{5} = \frac{12}{5} + \frac{4}{5} = \frac{16}{5}]$$

14. The Gradient Vector

Definition

The gradient of ($f(x,y)$) is the vector of partial derivatives:

$$[\nabla f = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \rangle = f_x \mathbf{i} + f_y \mathbf{j}]$$

For three variables:

$$[\nabla f = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k}]$$

Key Properties of the Gradient

Points in direction of steepest increase	The gradient tells you which way to go uphill fastest
$(\nabla f) = \text{rate of steepest increase}$	The magnitude is the maximum slope
$(-\nabla f) = \text{direction of steepest decrease}$	For gradient descent in optimization
Perpendicular to level curves	(∇f) is normal to the curve ($f = c$)

Maximum directional derivative:

$(\max D_{\mathbf{u}} f = \|\nabla f\|)$ (achieved when $(\mathbf{u})$ is in the direction of (∇f))

Minimum directional derivative:

$(\min D_{\mathbf{u}} f = -\|\nabla f\|)$ (in the direction of $(-\nabla f)$)

15. Algebra Rules for the Gradient

If (f) and (g) are differentiable functions and (c) is a constant:

Constant Multiple	$(\nabla(cf) = c\nabla f)$
Sum Rule	$(\nabla(f + g) = \nabla f + \nabla g)$
Difference Rule	$(\nabla(f - g) = \nabla f - \nabla g)$
Product Rule	$(\nabla(fg) = f\nabla g + g\nabla f)$
Quotient Rule	$(\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2})$
Chain Rule (scalar)	$(\nabla(g \circ f) = g'(f)\nabla f)$

Example (Product Rule):

$(\nabla(x^2 y^3) = y^3 \nabla(x^2) + x^2 \nabla(y^3) = \langle 2xy^3, 3x^2y^2 \rangle)$

Verify directly: $(f_x = 2xy^3), (f_y = 3x^2y^2) \checkmark$

16. Tangent Plane and Normal Line

Tangent Plane to a Surface

For the surface $(F(x,y,z) = 0)$, the tangent plane at point $(P_0 = (x_0, y_0, z_0))$ is:

$$[F_x(P_0)(x - x_0) + F_y(P_0)(y - y_0) + F_z(P_0)(z - z_0) = 0]$$

For the surface $(z = f(x,y))$, rewrite as $(F = f(x,y) - z = 0)$, so $(F_z = -1)$:

$$[z - z_0 = f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)]$$

Example:

Find the tangent plane to $(z = x^2 + y^2)$ at $((1, 1, 2))$.

$$[f_x = 2x = 2, \quad f_y = 2y = 2 \quad \text{at } (1,1)]$$

Tangent plane: $(z - 2 = 2(x-1) + 2(y-1) \Rightarrow z = 2x + 2y - 2)$

Normal Line

The normal line at (P_0) is

perpendicular to the tangent plane

. It runs in the direction of (∇F) :

$$[\mathbf{r}(t) = \langle x_0 + F_x t, y_0 + F_y t, z_0 + F_z t \rangle]$$

For $(z = f(x,y))$ at $((x_0, y_0, z_0))$:

$$[x = x_0 + f_x t, \quad y = y_0 + f_y t, \quad z = z_0 - t]$$

17. Second Derivative Test for Local Extreme Values

Critical Points

A critical point is where $(f_x = 0)$

and

$(f_y = 0)$ simultaneously. At these points, the tangent plane is horizontal.

The Discriminant (D-Test)

At a critical point $((a,b))$, compute the **discriminant**

:

$$[D = f_{xx}f_{yy} - (f_{xy})^2]$$

$(D > 0)$ and $(f_{xx} > 0)$	Local minimum
$(D > 0)$ and $(f_{xx} < 0)$	Local maximum
$(D < 0)$	Saddle point (neither max nor min)
$(D = 0)$	Inconclusive — test fails

Memory Trick:

"D positive means definite (max or min). D negative means diverge (saddle). D zero means dunno."

Example:

Find and classify the critical points of $(f(x,y) = x^3 + y^3 - 3x - 3y)$.

$$[f_x = 3x^2 - 3 = 0 \Rightarrow x = \pm 1] [f_y = 3y^2 - 3 = 0 \Rightarrow y = \pm 1]$$

Critical points: $((1,1), (1,-1), (-1,1), (-1,-1))$

$$[f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = 0] [D = 36xy]$$

$((1,1))$	$(36 > 0)$	$(6 > 0)$	Local minimum
$((-1,-1))$	$(36 > 0)$	$(-6 < 0)$	Local maximum
$((1,-1))$	$(-36 < 0)$	—	Saddle point
$((-1,1))$	$(-36 < 0)$	—	Saddle point

PART III: MULTIPLE INTEGRALS

18. Double Integrals over a Rectangle

Setup

A double integral over a rectangular region ($R = [a,b] \times [c,d]$) is:

$$\iint_R f(x,y) \, dA = \int_c^d \int_a^b f(x,y) \, dx \, dy$$

Geometric Meaning: If ($f(x,y) \geq 0$), the double integral gives the **volume** under the surface ($z = f(x,y)$) above the rectangle (R).

19. Fubini's Theorem (First Form)

Statement

If ($f(x,y)$) is continuous on the rectangle ($R = [a,b] \times [c,d]$), then:

$$\iint_R f(x,y) \, dA = \int_c^d \int_a^b f(x,y) \, dx \, dy = \int_a^b \int_c^d f(x,y) \, dy \, dx$$

The Big Idea: You can compute a double integral as two successive single integrals, and **the order doesn't matter** (for continuous functions on rectangles). This is Fubini's Theorem.

Example:

Evaluate ($\iint_R xy \, dA$) where ($R = [0,2] \times [0,3]$).

$$\int_0^3 \int_0^2 xy \, dx \, dy = \int_0^3 \left[\frac{x^2 y}{2} \right]_0^2 dy = \int_0^3 2y \, dy = \left[y^2 \right]_0^3 = 9$$

20. Fubini's Theorem (Strong Form)

For Non-Rectangular Regions

The strong form handles regions bounded by curves:

Type I Region (bounded by functions of (x)): $\left[\iint_R f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y) \, dy \, dx \right]$

Type II Region (bounded by functions of (y)): $\left[\iint_R f \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y) \, dx \, dy \right]$

Example:

Evaluate $\left(\iint_R y \, dA \right)$ where (R) is the region bounded by $(y = x)$ and $(y = x^2)$ for $(0 \leq x \leq 1)$.

$$\begin{aligned} \left[\int_0^1 \int_{x^2}^x y \, dy \, dx &= \int_0^1 \left[\frac{y^2}{2} \right]_{x^2}^x dx = \int_0^1 \frac{x^2 - x^4}{2} \, dx = \frac{1}{2} \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \frac{1}{2} \cdot \frac{2}{15} = \frac{1}{15} \right] \end{aligned}$$

Reversing the Order of Integration

Sometimes one order is easier to evaluate than the other. Sketch the region, then rewrite the limits for the reversed order.

21. Average Value of f over a Region R

The average value of $(f(x,y))$ over a region (R) with area (A):

$$\left[\bar{f} = \frac{1}{A} \iint_R f(x,y) \, dA \right]$$

Example:

Average value of $(f(x,y) = x + y)$ over $(R = [0,2] \times [0,2])$ (area = 4):

$$\left[\iint_R (x+y) \, dA = \int_0^2 \int_0^2 (x+y) \, dx \, dy = \int_0^2 [2x + yx]_0^2 \, dy = \int_0^2 (4 + 2y) \, dy = [4y + y^2]_0^2 = 8 \right] \left[\bar{f} = \frac{8}{4} = 2 \right]$$

(Makes sense — the average of $(x + y)$ over $([0,2]^2)$ should be $(1 + 1 = 2)$.)

22. Double Integrals in Polar Coordinates

Why Use Polar Coordinates?

For regions shaped like circles, sectors, or rings, polar coordinates simplify the integral significantly.

Recall: $(x = r\cos\theta)$, $(y = r\sin\theta)$, and the crucial factor:

$$[dA = r\,dr\,d\theta]$$

Do not forget the extra (r)!

This comes from the Jacobian of the transformation.

$$[\iint_R f(x,y)\,dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\,dr\,d\theta]$$

Common polar regions:

Full disk of radius (a)	$(0 \leq r \leq a), (0 \leq \theta \leq 2\pi)$
Upper half-disk	$(0 \leq r \leq a), (0 \leq \theta \leq \pi)$
Annulus (ring)	$(a \leq r \leq b), (0 \leq \theta \leq 2\pi)$
Sector	$(0 \leq r \leq a), (\alpha \leq \theta \leq \beta)$

Example:

Evaluate $(\iint_R e^{-(x^2+y^2)}\,dA)$ over the disk $(x^2 + y^2 \leq 4)$.

$$[\int_0^{2\pi} \int_0^2 e^{-r^2} \cdot r\,dr\,d\theta = 2\pi \int_0^2 r e^{-r^2}\,dr = 2\pi \left[-\frac{1}{2} e^{-r^2} \right]_0^2 = \pi(1 - e^{-4})]$$

23. Triple Integrals in Rectangular Coordinates

Extension to 3D

For a function $(f(x,y,z))$ over a 3D region (D) :

$$[\iiint_D f(x,y,z)\,dV = \int_a^b \int_c^d \int_e^g f(x,y,z)\,dz\,dy\,dx]$$

Again, Fubini's theorem says we can change the order of integration (6 possible orders for three variables).

Step-by-step approach:

1. Sketch the region (D) 2. Identify the outermost limits (constants) 3. Middle limits depend on the outermost variable 4. Innermost limits depend on both outer variables

Example:

$(\iiint_D x \, dV)$ where (D) is the box $([0,1] \times [0,2] \times [0,3])$.

$$\left[\int_0^1 \int_0^2 \int_0^3 x \, dz \, dy \, dx = \int_0^1 \int_0^2 3x \, dy \, dx = \int_0^1 6x \, dx = 3 \right]$$

24. Volume of a Closed Bounded Region in Space

If $(f(x,y,z) = 1)$, the triple integral gives the

volume

:

$$\left[V = \iiint_D dV \right]$$

Example:

Find the volume of the region bounded by $(z = 0)$, $(z = 4 - x^2 - y^2)$ (a paraboloid).

In cylindrical coordinates (easier — see Section 26), $(x^2 + y^2 = r^2)$:

$$\left[V = \int_0^{2\pi} \int_0^2 \int_0^{4-r^2} r \, dz \, dr \, d\theta = 2\pi \int_0^2 r(4-r^2) \, dr = 2\pi \left[2r^2 - \frac{r^4}{4} \right]_0^2 = 2\pi(8-4) = 8\pi \right]$$

25. Average Value of a Function in Space

The average value of $(f(x,y,z))$ over region (D) with volume (V):

$$\left[\bar{f} = \frac{1}{V} \iiint_D f(x,y,z) \, dV \right]$$

Example:

Average temperature $(T(x,y,z) = x + y + z)$ over the unit cube $([0,1]^3)$:

$$\left[\iiint T \, dV = \int_0^1 \int_0^1 \int_0^1 (x+y+z) \, dx \, dy \, dz = \frac{3}{2} \right] \left[\bar{T} = \frac{3/2}{1} = \frac{3}{2} \right]$$

(Makes sense: average of each of (x) , (y) , (z) over $([0,1])$ is $(1/2)$, and average of their sum is $(3/2)$.)

26. Triple Integrals in Cylindrical and Spherical Coordinates

Cylindrical Coordinates

Transformation:

$$[x = r\cos\theta, \quad y = r\sin\theta, \quad z = z] \quad [dV = r\,dr\,d\theta\,dz]$$

Use when:

The region or integrand involves $(x^2 + y^2)$ (circles in (xy) -plane), or has cylindrical symmetry.

Example:

Volume of the cylinder $(x^2 + y^2 \leq 9)$, $(0 \leq z \leq 5)$:

$$\int_0^{2\pi} \int_0^3 \int_0^5 r\,dz\,dr\,d\theta = 2\pi \cdot \int_0^3 5r\,dr = 2\pi \cdot \frac{45}{2} = 45\pi$$

Verify: $(V = \pi r^2 h = \pi(9)(5) = 45\pi) \checkmark$

Spherical Coordinates

Transformation:

$$[x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi]$$

where $(\rho \geq 0)$ is the distance from origin, $(\phi \in [0, \pi])$ is the angle from the positive z -axis, and $(\theta \in [0, 2\pi])$ is the angle in the xy -plane.

$$[dV = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta]$$

The factor $(\rho^2\sin\phi)$ must not be forgotten!

Use when:

The region is a sphere, cone, or has spherical symmetry. Expressions like $(x^2 + y^2 + z^2 = \rho^2)$ simplify beautifully.

Key identities:

$$[x^2 + y^2 + z^2 = \rho^2] \quad [x^2 + y^2 = \rho^2\sin^2\phi]$$

Comparison of Coordinate Systems

(dV)	(dx, dy, dz)	(r, dr, dθ, dz)	(ρ ² sin φ, dρ, dφ, dθ)
Best for	Boxes, general regions	Cylinders, paraboloids	Spheres, cones
Key expression	—	(x ² + y ² = r ²)	(x ² + y ² + z ² = ρ ²)

Example (Sphere volume):

Volume of sphere (ρ ≤ a):

$$\int_0^{2\pi} \int_0^{\pi} \int_0^a \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 2\pi \cdot \frac{a^3}{3} \cdot \int_0^{\pi} \sin \phi \, d\phi = 2\pi \cdot \frac{a^3}{3} \cdot 2 = \frac{4}{3}\pi a^3 \quad \checkmark$$

APPENDIX: Quick Reference Formulas

Limits

Direct substitution	Polynomial/continuous functions
Two-path test	Prove limit does NOT exist
Iterated limits	Prove limit does NOT exist
Polar coordinates	Limits as ((x,y)→(0,0))

Derivatives Summary

Partial derivative (f_x)	Differentiate w.r.t. (x), treat (y) as constant
Total differential	$(df = f_x dx + f_y dy + f_z dz)$
Directional derivative	$(D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u})$
Gradient	$(\nabla f = \langle f_x, f_y, f_z \rangle)$
Tangent plane	$(z - z_0 = f_x(x - x_0) + f_y(y - y_0))$
2nd Derivative Test	$(D = f_{xx}f_{yy} - f_{xy}^2)$
Laplace Equation	$(f_{xx} + f_{yy} + f_{zz} = 0)$

Integrals Summary

Rectangular ((x,y,z))	$(dV = dx dy dz)$
Polar ((r,theta))	$(dA = r dr d\theta)$
Cylindrical ((r,theta,z))	$(dV = r dr d\theta dz)$
Spherical ((rho,phi,theta))	$(dV = \rho^2 \sin\phi d\rho d\phi d\theta)$

Document prepared for Multivariable Calculus students. Covers Limits, Partial Derivatives, and Multiple Integrals with worked examples.